Can LLMs Simulate Travellers?

Comparing a local instruct model with a multinomial logit on Swissmetro

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Question

Mode-choice models used in appraisal are fitted to surveys. A recent suggestion is that a language model can take the respondent's place: describe the person and the alternatives, and treat the model's reply as a simulated choice. I tested that suggestion on one public stated-preference dataset and one small open-weights model, against the multinomial logit that the same dataset is usually taught with.

The dataset is Swissmetro (Bierlaire, Axhausen, and Abay, 2001). Respondents chose among the existing train, a proposed maglev, and a car. I kept the standard analysis sample: trip purpose commute or business, and a stated choice. That is 6,768 tasks from 752 people, nine tasks each. For holders of the General Abonnement, the raw train "cost" is the annual pass price. I set the train and Swissmetro trip costs to zero for those people, which is what the Biogeme example does, and I told the language model the same thing in words.

The logit

The baseline is the Biogeme specification `b01a`. Swissmetro's alternative-specific constant is fixed at zero. One time coefficient and one cost coefficient are shared. Time and cost are divided by 100. I maximise the logit log-likelihood by Newton steps and report cluster-robust standard errors, with one cluster per person, because nine answers from one person are not nine people.

On the full sample the estimates match the published example: log-likelihood -5,331.252 against a null of -6,964.663, $\rho^2 = 0.235$, $\bar{\rho}^2 = 0.234$. The coefficients are ASC train -0.701 (cluster t = -3.82), ASC car -0.155 (cluster t = -1.20), time -1.278 (cluster t = -5.38), and cost -1.084 (cluster t = -6.72). The car constant is the parameter that looks significant when each task is treated as independent (robust t = -2.66) and does not survive clustering. The value of time is 60 times the time coefficient over the cost coefficient: 70.7 CHF per hour, with a cluster standard error of 13.8. That number belongs to this stated-preference design. I would not carry it into an appraisal.

A richer model adds business-trip shifts of the train and car constants, pass-holder shifts, headway, an extra time coefficient for age 55 and over, and an extra cost coefficient for income above 100,000 CHF per year. The log-likelihood rises to -4,889.826 ($\bar{\rho}^2 = 0.296$). The likelihood-ratio statistic against the base model is 882.85 on 7 degrees of freedom, but that test counts tasks as independent. Under clustering, the pass shift on Swissmetro (t = 0.15) and the high-income cost shift (t = 1.35) are weak. The pass shift on the train (t = 4.39), headway (t = -5.84), and the senior time shift (t = 3.04) are not. For a non-senior person outside the top income band the richer value of time is 64.8 CHF/hour. For age 55+ it is 32.1. The high-income point estimate is 96.1 CHF/hour and I do not treat it as established. The train pass shift is worth 153.8 CHF on top of a zero fare (s.e. 55.5). People who already prefer the train are those who buy the pass, so this is not a causal effect of the pass.

I held out 150 people (1,350 tasks) by respondent ID, seed 42, and estimated on the other 602. On

the held-out people the base model is accurate on 64.3% of tasks, the richer model on 67.4%, and a constants-only model on 57.3%, which is also what a majority-class rule achieves by always picking Swissmetro. Mean predicted probabilities from the base model are 13.4% train, 57.9% Swissmetro, and 28.7% car, against observed shares of 13.9%, 57.3%, and 28.8%. The hard prediction of the base model never selects the train. Accuracy is the wrong single summary of that model. The test log-likelihood is -1,138.100 for the base model and -1,060.671 for the richer one.

The language model

The model is Qwen2.5-1.5B-Instruct, stored as a 4-bit GGUF (`Q4_K_M`) and run locally with llama.cpp. No API, and no sampling. The probability of each mode is the next-token probability of the digit 1, 2, or 3, renormalised over the modes that are available. The prompt, which is in the repository, gives purpose, age, gender, income, who pays, luggage, the annual pass, and the time, cost, and headway of each option. I checked the format on 24 training respondents (every top token was a digit) and did not edit the wording to chase accuracy. The scored sample is 420 tasks from 147 held-out people. Elasticities use 84 of those tasks. The run is deterministic given the weights and the prompt files.

Comparison

On the 420 tasks the majority rule scores 57.4%. The base logit scores 63.1% (bootstrap interval 56.8 to 69.6, resampling the 147 people). The richer logit scores 67.1%. The language model scores 34.3% (interval 28.1 to 40.7). The gap between the base logit and the language model is 28.8 percentage points, with an interval from 18.8 to 39.5. The language model is 23.1 points below the majority rule, and that interval stays below zero.

Log-likelihood on the same tasks tells the same story more sharply: -357.22 for the base logit, -324.01 for the richer logit, -381.03 for constants only, and -693.20 for the language model. Its mean probability on the chosen alternative is 0.344.

The shares explain the accuracy. Observed shares are 13.8% train, 57.4% Swissmetro, 28.8% car. Base-logit probabilities are 13.4 / 58.0 / 28.7. Language-model probabilities are 29.6 / 12.7 / 57.7, and its hard prediction is 20.0 / 1.4 / 78.6. It selects Swissmetro on 6 of 420 tasks and is right on 2 of the 241 Swissmetro choices. It is comparatively good at the train (recall 51.7%), which is the mode the base logit's hard rule never picks. The two hard predictions agree on 16.0% of tasks. Task-level probabilities correlate 0.38 for the train, -0.04 for Swissmetro, and 0.36 for the car.

The annual pass is the place where the language model tracks the sample and the base logit does not. On 41 pass-holder tasks the observed train share is 73.2% and the model's mean train probability is 78.3%. On the other 379 tasks those figures are 7.4% and 24.3%. The base logit, which only sees a zero fare, stays near 16.6% and 13.0%. Removing the sentence that names the pass, and leaving the fares at zero, moves the train probability on those 41 tasks from 78.3% to 60.1%. Both the zeros and the wording matter. Sixteen people is a thin sample for a finer claim.

The time and cost shocks go the other way. All 24 base-logit own elasticities for train and Swissmetro time and cost are negative, and each bootstrap interval excludes zero. Making the train 50% slower moves the logit train probability from 12.5% to 5.4%. It moves the language-model probability from 28.0% to 30.1% (elasticity +0.16, interval 0.04 to 0.31). Making the train 50% faster also lowers the language-model train probability. A +/-10 minute and +/-10 CHF central difference recovers 72.6 CHF/hour from the logit, matching the training-sample coefficient ratio (cluster s.e.

16.2). The same shifts move the language-model share in the wrong direction, so I do not report a language-model value of time. One Swissmetro fare increase of 50%, on 76 fare-paying tasks, has a correctly signed language-model elasticity whose interval excludes zero (-0.16, interval -0.27 to -0.04). The probability only moves from 14.0% to 12.9%. That single cell does not describe the cost responses as a whole.

When time, cost, and headway are equalised on the 84-task shock sample, the logit still puts 44.7% on Swissmetro and the language model puts 46.7% on the car. Swissmetro is the fast alternative in the raw file (mean 84.5 minutes, against 166.1 by train). The model under-chooses it after that advantage is removed. The bias is a car prior, not a blindness to the annual pass.

What this does not show

It does not show that language models cannot represent travellers. It shows that this 1.5-billion-parameter, 4-bit model, with this one prompt, on 420 stated-preference tasks, does not. The logit was fitted to 602 other people from the same survey. The language model was not. A larger model, or a fine-tune on the training respondents, is a different experiment, and I have not run it.

The data are stated choices about a mode that was never built, from a sample dominated by business travellers and by men. The elasticity subsample is 84 tasks. The pass-wording check is 16 people. Panel dependence is handled with cluster standard errors and a respondent bootstrap; tests that treat tasks as independent, including the likelihood-ratio p-value and the McNemar tests stored in the output, overstate precision. The base logit's hard predictions hide a well calibrated train probability. I would quote the log-likelihood and the probability shares in any interview about this repository, and I would quote 34.3% against 63.1% against 57.4% as the accuracy line, with the bootstrap intervals, and with the car share as the reason.

The files `results/summary.json`, `report.md`, and `docs/explainer.md` are three views of one run. If they ever disagree, the JSON is the one the code wrote.